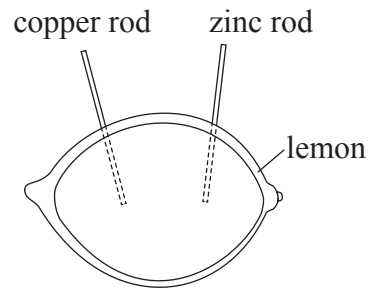


(Question 6, part 1 continued)

Part 2 Electricity

- (d) A lemon can be used to make an electric cell by pushing a copper rod and a zinc rod into the lemon.



A student constructs a lemon cell and connects it in an electrical circuit with a variable resistor. The student measures the potential difference V across the lemon and the current I in the lemon.

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(Question 6, part 2 continued)

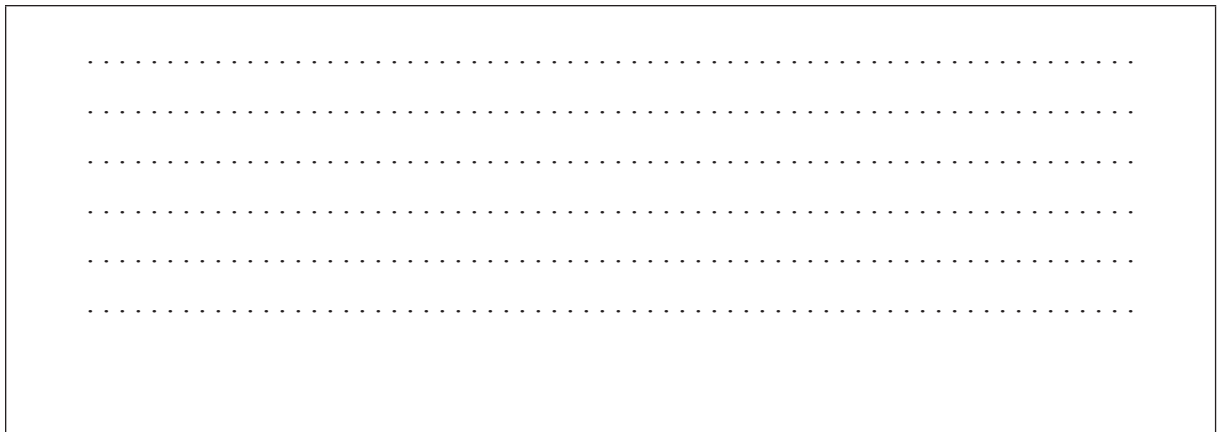
- (i) Draw a circuit diagram of the experimental arrangement that will enable the student to collect the data for the graph. [2]



- (ii) Show that the potential difference V across the lemon is given by

$$V = E - Ir$$

where E is the emf of the lemon cell and r is the internal resistance of the lemon cell. [2]

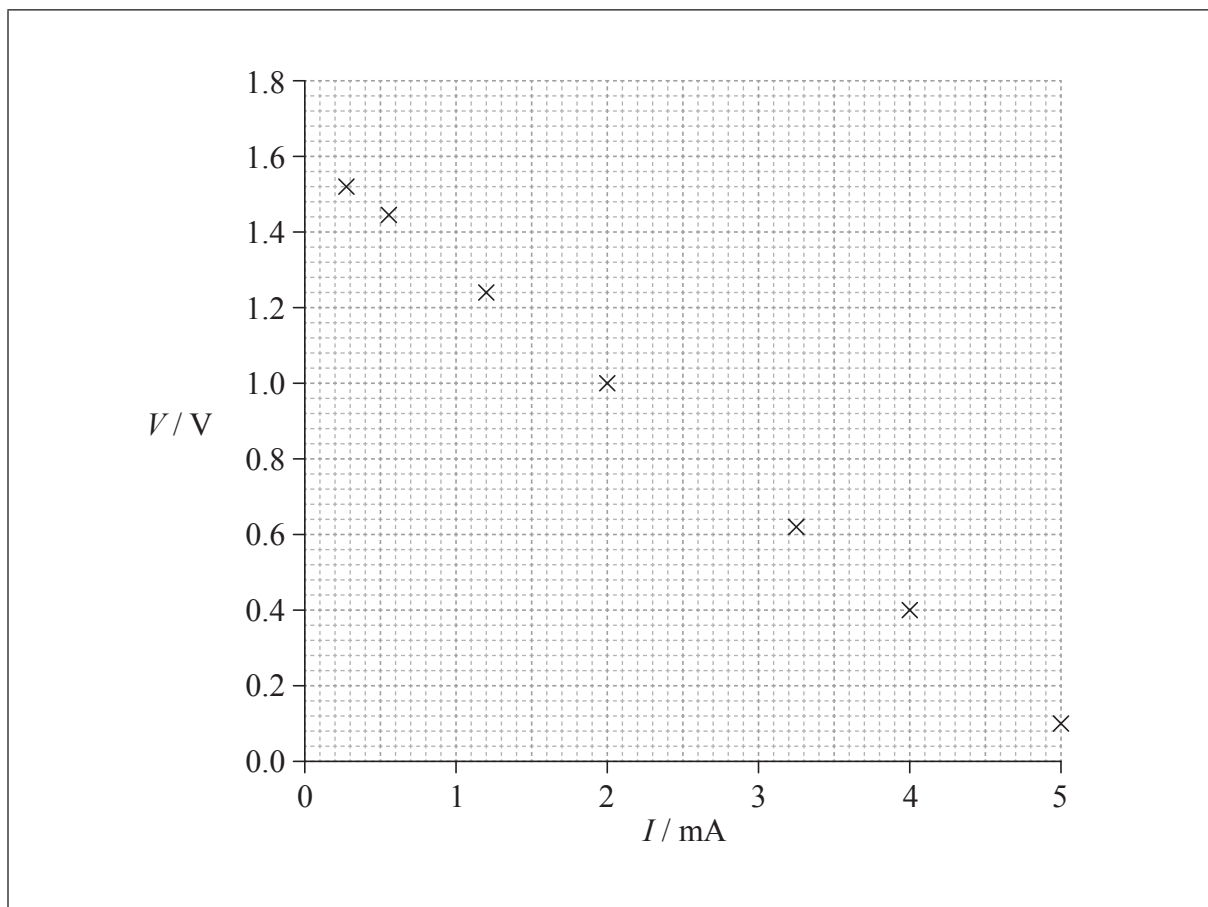


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(Question 6, part 2 continued)

(iii) The graph shows how V varies with I .



Using the graph, estimate the emf of the lemon cell.

[2]

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(Question 6, part 2 continued)

- (iv) Determine the internal resistance of the lemon cell.

[3]

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- (v) The lemon cell is used to supply energy to a digital clock that requires a current of $6.0\mu\text{A}$. The clock runs for 16 hours. Calculate the charge that flows through the clock in this time.

[1]

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